



AUTOMATE INDIA 2026 — NIET CHAPTER

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AUTOMATE
INDIA 2026

PROJECT CIRCULARCHAIN

Autonomous Circular Economy,
CPCB Statutory EPR Compliance &
On-Chain Carbon Ledger Protocol



01



Multi-Modal AI Sentinel

Real-time visual scrap
purity grading &
contamination heatmaps.

02



CPCB Compliance Matrix

FY26-27 statutory EPR
quotas, SPCB hub rules &
Form 1 audit export.

03



Polygon Amoy Ledger

ERC-1155 verifiable batch
escrow & live Indian MCX
commodity spot oracle.

PRESENTED BY:
Team Rocket



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TEAM MEMBERS



Nishant Gaurav

TEAM LEAD & PRINCIPAL SYSTEM ARCHITECT

Full-Stack System Architect | Web3 & Blockchain Lead | AI Vision Core



Rudra Pratap Nagar

AI / ML INTELLIGENCE & COMPUTER VISION SENTINEL

AI & ML Specialist | Computer Vision Engineer | Data & Model Analyst



Neeraj Kumar

FULL-STACK SYSTEMS & STATUTORY COMPLIANCE

Full-Stack Developer | Systems Engineer | Compliance & Integration

PROBLEM STATEMENT

Autonomous Circular Economy & Carbon Ledger Protocol for Industrial Scrap & CPCB Compliance.

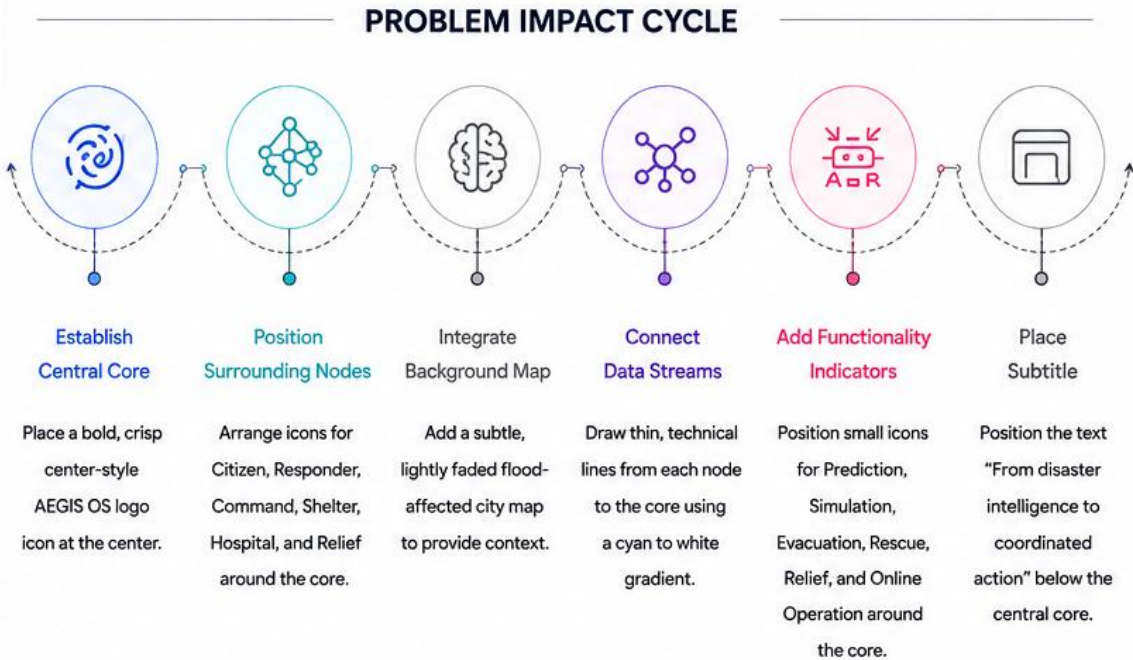
WHAT DOES THIS PROBLEM AIM TO SOLVE?

India generates over **62 Million Tonnes** of industrial and plastic waste annually. Despite strict government EPR (Extended Producer Responsibility) mandates, 70%+ of scrap transactions operate in untraceable black markets riddled with **fake invoices, ghost recycling claims, and severe contamination**.

Brand owners face heavy CPCB penalties (up to **₹15,000/MT** under EPA Section 15) because they cannot verify if their scrap was actually recycled, while informal waste pickers (Kabadiwalas) are exploited without fair market pricing.

WHY DID YOUR TEAM CHOOSE THIS?

We chose this because we wanted to build one unified autonomous operating system that connects industrial waste generators, certified recyclers, and government regulators. By combining AI vision inspection with blockchain smart contracts, we eliminate greenwashing and make recycling **100% mathematically verifiable**.



1

FRAUD & GHOST RECYCLING
Fake recycling certificates and paper invoices lead to massive greenwashing and non-compliance fines.

2

COMPLEX CPCB MATH
Brand owners struggle to calculate statutory quotas across 8+ material classes and risk ₹15,000/MT penalties.

3

CONTAMINATION & FAIR PRICE
Recyclers reject contaminated scrap batches, while grassroots waste pickers lack live MCX price transparency.

REAL-WORLD PROBLEM ALIGNMENT & POLICY INTEGRATION

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What is the Current Gap in the System?



1. Disconnected State & Central Portals:

CPCB and state SPCBs (UPPCB, MPCB, GPCB) operate as siloed PDF submission forms with zero live verification.



2. Retrospective Audits:

Inspections happen 6 months after disposal, making it impossible to detect fictitious recycling lots.



3. Informal Sector Exploitation:

1.5M+ waste pickers lack live fair-market price discovery.



4. Opaque Carbon Accounting:

No cryptographic linkage between recycled metric tonnes and verified Scope 3 CO₂e emissions reduction.



Alignment with Indian Government Schemes



1. MoEFCC Plastic Waste Management Rules 2026-27:

Directly executes statutory EPR quotas (Cat I 80%, Cat II 70%) and mandatory PCR recycled plastic blend quotas.



2. CPCB Form 1 & EPA Section 15 Compliance:

One-click export of statutory Form 1 audit reports and automated calculation of avoided environmental compensation penalties (₹15,000/MT).



3. Mission LiFE (Lifestyle for Environment):

Enables transparent circular adoption for FMCG & manufacturing giants with verified resource tracking.



4. MCX Commodity Spot Benchmark Linkage:

Integrates National Multi-Commodity Exchange spot indexes for scrap aluminum, copper, PET, and steel fair pricing.

PROPOSED SOLUTION — 7-STEP AUTONOMOUS LIFECYCLE



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We turn scrap into impact.

Our platform digitizes and verifies every scrap lot, converts it into carbon value, and settles everything on-chain — no paperwork, no manual follow-ups, just real-time, verifiable results.

We act before problems happen.

Using live data, AI vision, and market intelligence, we trigger the right actions instantly — accurate pricing, routing, compliance, and payments without any human delay.

We adapt to every change.

Whether it's price fluctuation, route congestion, or compliance updates — our system recalibrates automatically to ensure smooth operations, lower emissions, and continuous value for all stakeholders.



1. Autonomous Material Passport

Digital twin with immutable purity, lot origin, and carbon credentials.



2. Multi-Hub SPCB Engine

Full support for SPCB regional hubs (UPPCB, MPCB, GPCB) with PCR mandates.



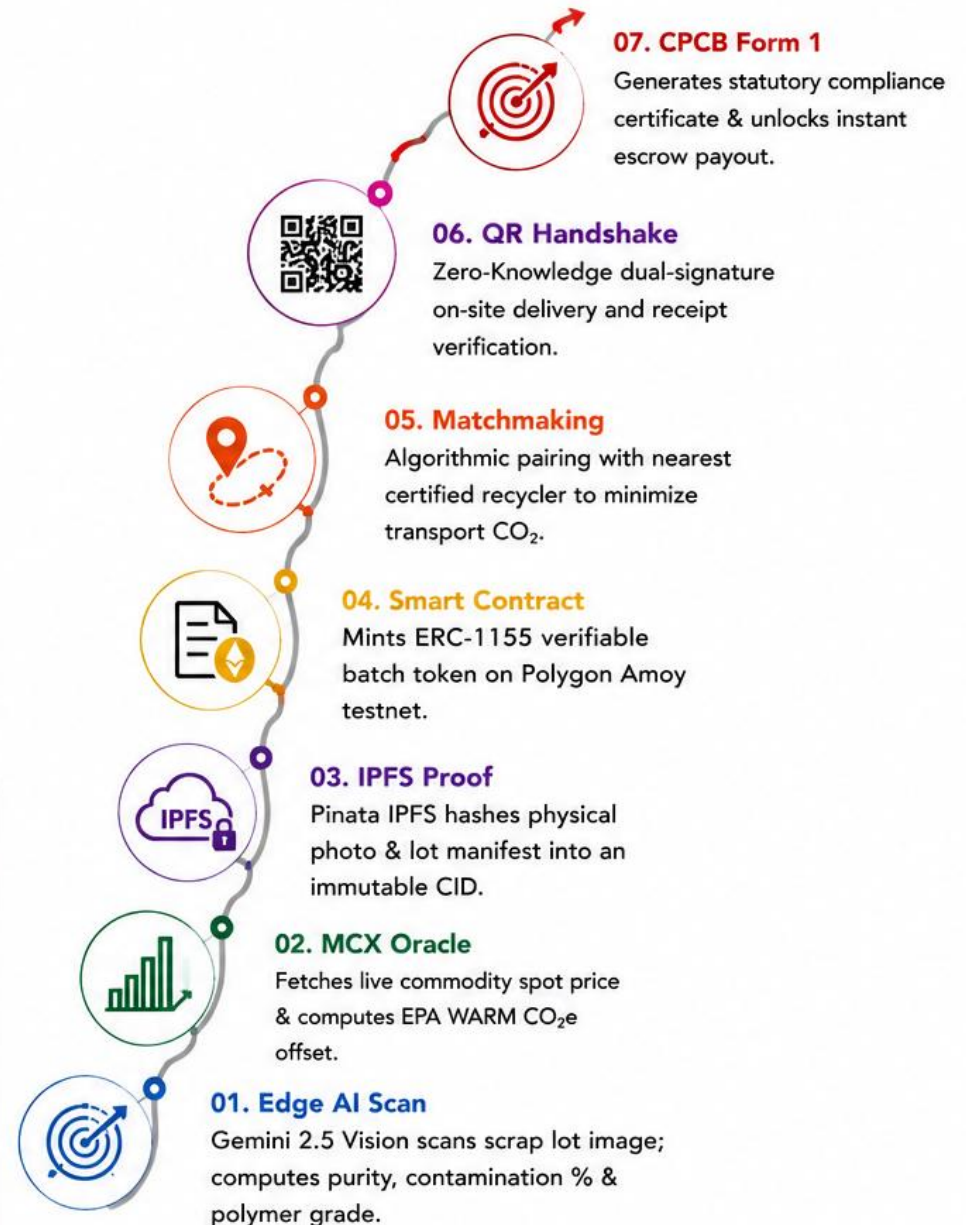
3. On-Chain Carbon Ledger

Polygon Amoy tokenized certificates eliminating double-counting.



4. Indic Voice Assistant

Multilingual voice interaction in Hindi/Tamil for grassroots scrap yard operators.



INNOVATION & COMPETITIVE DIFFERENTIATORS



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01

Multi-Modal AI Contamination Vision

Unlike existing platforms that rely on unverified seller self-declarations, our Gemini 2.5 Vision pipeline analyzes lot photographs in real time, detecting micro-contaminants, moisture, composite layers, and calculates accurate purity scores.



02

Live Indian MCX Commodity Oracle

Prevents price manipulation and exploitation of informal waste pickers by pulling live scrap spot prices across Aluminum 6063, Copper Berry, PET Flakes, HDPE Granules, and HMS Steel directly from Indian commodity benchmarks.



03

Cryptographic Zero-Knowledge Transfer Proof

Solves the ₹4,500 Cr ghost recycling scam. Materials can only be marked as recycled when both Generator and Recycler perform a dual-signature QR verification registered on the Polygon Amoy blockchain ledger.



04

SPCB Statutory Audit & Indic Voice Interface

Automates complex Form 1 audit filings, regional SPCB compliance rules (UPPCB, MPCB, GPCB), and integrates multi-dialect Indic voice assistance so grassroots yard workers can operate seamlessly.



End-to-End Autonomous Lifecycle

From scrap lot scan to compliance payout — fully automated, traceable & on-chain.

04. SPCB Statutory Audit & Indic Voice Interface

Automated compliance & multi-dialect Indic voice assistance.

01. Multi-Modal AI Contamination Vision

Real-time AI analysis for accurate purity & contamination detection.

02. Live Indian MCX Commodity Oracle

Live scrap prices from Indian commodity benchmarks.

03. Cryptographic Zero-Knowledge Transfer Proof

Dual-signature QR verification on Polygon Amoy blockchain prevents fraud.

CIRCULARCHAIN

Transparent • Traceable • Trusted

TECHNICAL STACK & INTEGRATED APIS



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01 Frontend & Mobile



Next.js 14
(App Router & TypeScript)



Flutter 3.41+ (Dart)
Cross-Platform



Tailwind CSS
(Onyx Dark & Light Studio)



Lucide Icons &
Framer Motion



Android Studio
Production Emulator

02 Backend & APIs



Node.js & Express
(TypeScript)



Prisma ORM &
PostgreSQL / SQLite



Unified RESTful
Data Layer
(Port 5000)



Pinata IPFS
Gateway Client



Multi-Tenant
Organization Directory

03 AI & Computer Vision



Google Gemini 2.5
Multi-Modal Vision



Automated
Contamination
Bounding Heatmaps



PCR Blend Quality
Classifier



US EPA WARM
Emission Engine



Indic Speech &
Voice Processing

04 Blockchain & Web3



Solidity ERC-1155
Multi-Token Protocol



Polygon Amoy Testnet
(Chain ID 80002)



Ethers.js v6
Transaction Suite



IPFS Content
Addressing



Dual-Signature
Escrow Lock

05 Government & Oracles



CPCB Statutory
Matrix FY26-27



SPCB Regional
Multi-Hub Rules
(UPPCB/MPCB)



MCX Spot Scrap
Price Oracle Feed



EPA Section 15
Penalty Estimator



Automated Form 1
Certificate Generator

DEVELOPMENT PLAN & LONG-TERM VISION

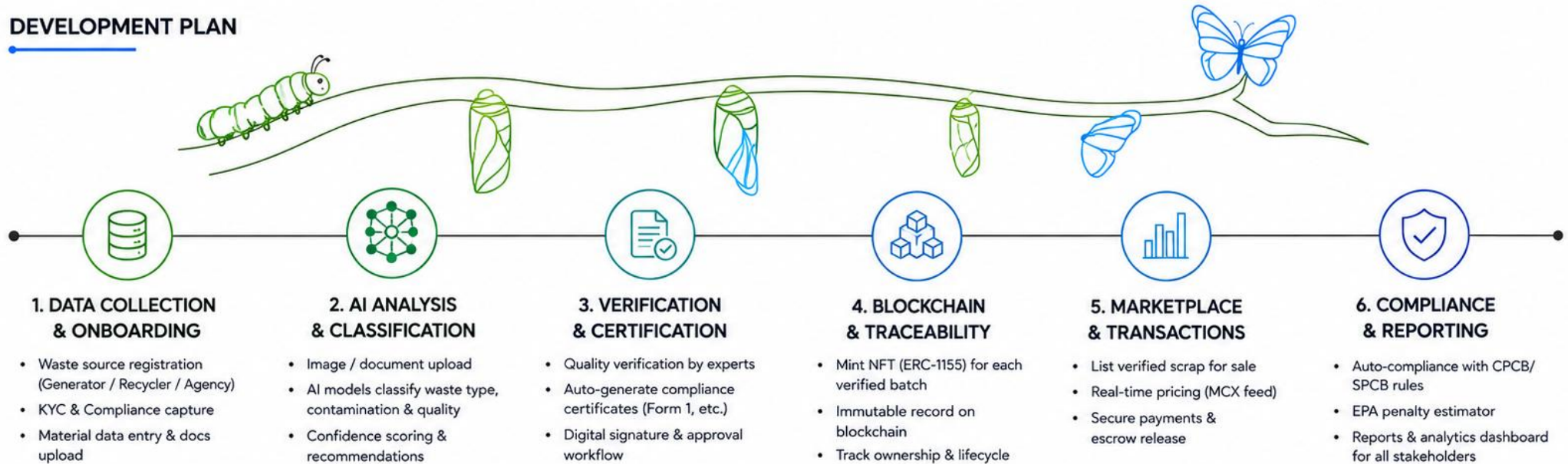


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DEVELOPMENT PLAN



LONG-TERM SUSTAINABILITY



HIGH-LEVEL SYSTEM ARCHITECTURE



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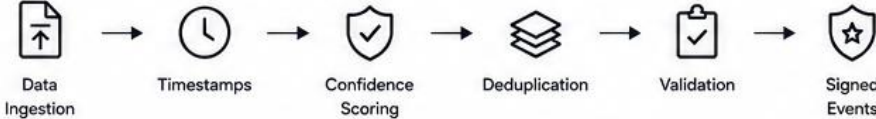


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EXTERNAL / AUTHORIZED INPUTS



INGESTION + TRUST LAYER



CORE SHARED STATE

CircularChain Digital Twin



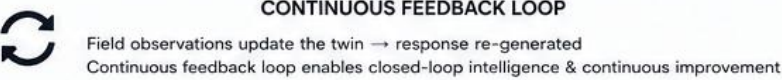
INTELLIGENCE & DECISION ENGINES



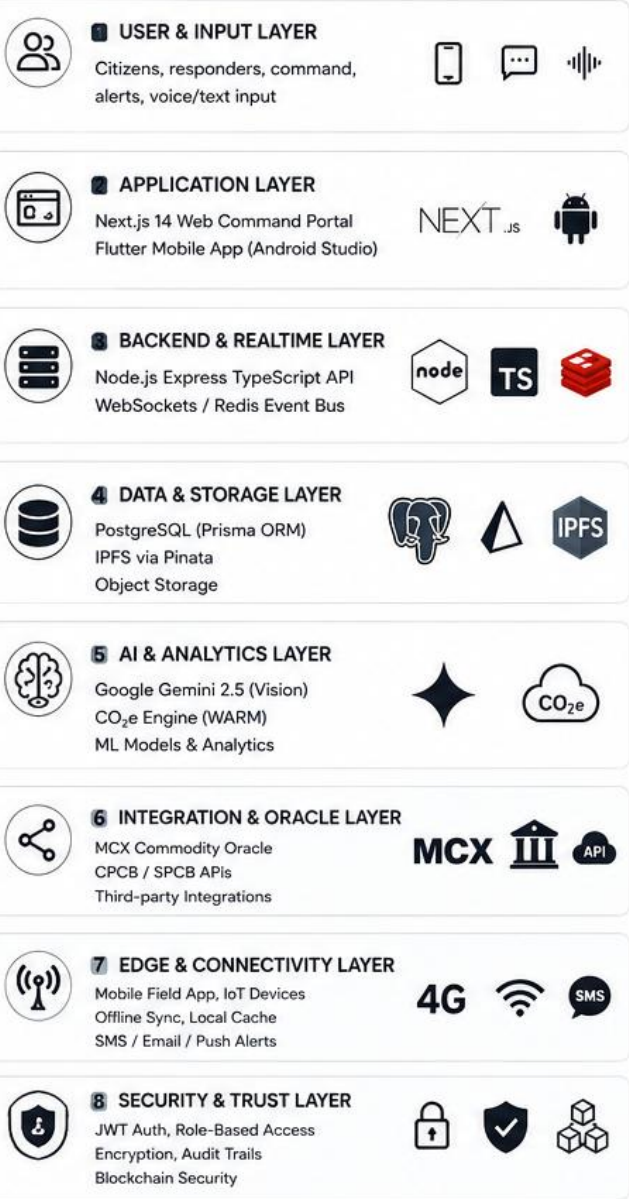
ROLE-BASED OPERATING SURFACES / APPLICATIONS



CONTINUOUS FEEDBACK LOOP



TECH STACK LAYERS



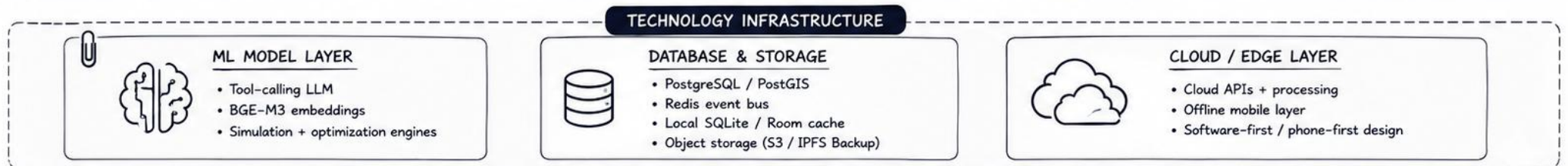
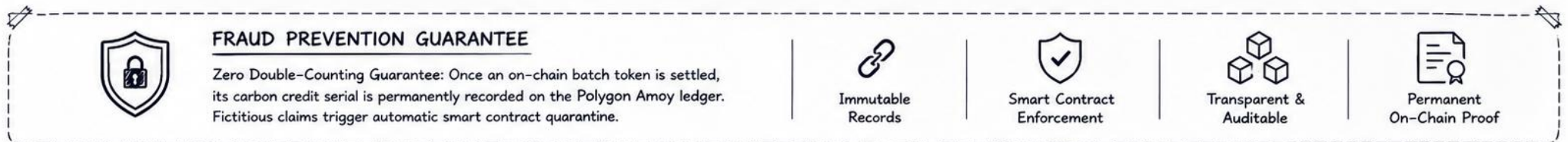
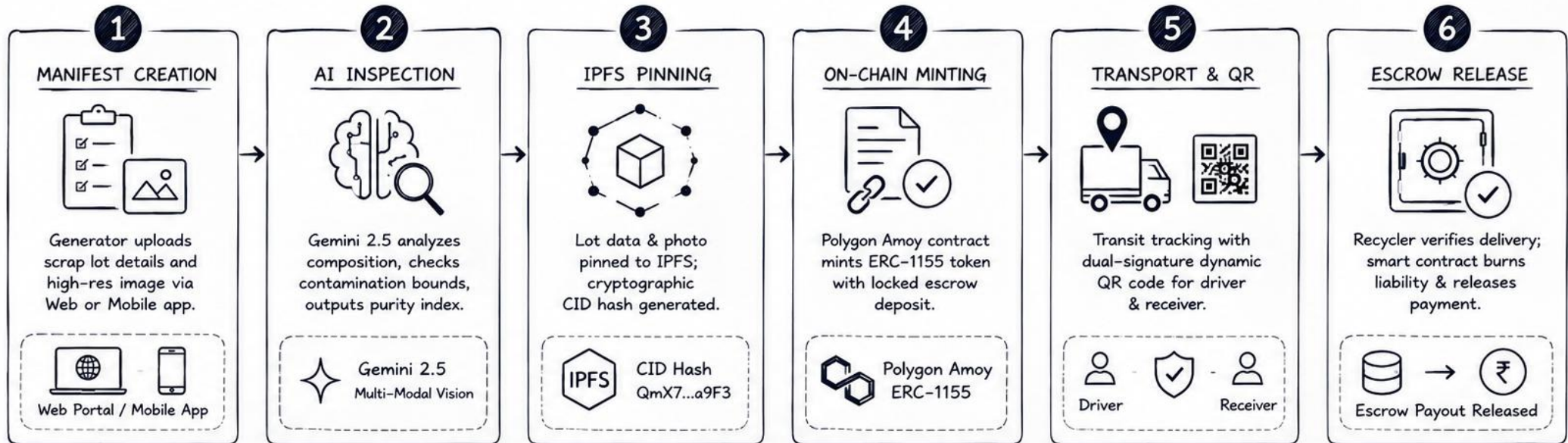
TECHNICAL DATA FLOW & VERIFICATION PIPELINE



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CROSS-PLATFORM PRODUCTION UI SHOWCASE



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Unified Digital Twin Experience Across Web Command Center & Mobile Field App



WEB COMMAND CENTER (Next.js 14)

CircularChain

Dashboard

Marketplace

EPR Calculator

AI Inspection

Blockchain Explorer

Reports

Compliance

Settings

Dashboard Overview

Total Scrap Listed

12,450

+10.4% this week

Avg Purity Index

92.4%

+4.1% this week

CO₂e Abated

1,245 t

+21.7% this week

Active Transactions

320

+12.7% this week

Circular Scrap Marketplace

Search scrap type, material, location...

Purity: 80%+ More Filters

Material	Purity (%)	Carbon	Carbon (kg/t)	Time
Aluminum Scrap (Taint Tabor)	95.6%	2.4	Delhi, DL	2m ago
Steel Scrap (Shredded)	91.2%	1.8	Mumbai, MH	5m ago
Copper Scrap (Mixed)	93.1%	3.2	Pune, MH	8m ago

EPR Calculator (CPCB)

State Matrix

UPPCB

Material

Steel

Quantity (MT)

100

EPR Liability (₹)

12,450

CO₂e Impact (t)

2.16

Download Form 1 (PDF)

View Regional Matrix

AI Contamination Heatmap

Polygon Amoy Explorer

TX Hash

0x8f3a...7c9e

Verified

Status

Confirmed

Block

51234567

Time

2 mins ago

View on Explorer

Government Export

Form 1 - CPCB

PDF

JSON Export

Download

Export History

Amrit Verma

Admin

MOBILE FIELD APP (Flutter / Android Studio)

Dashboard

Hi, Operator

Greenfield Scrap Yard

Quick Overview

Purity Index

92.4%

+4.1%

CO₂ Savings

2.48 t

+12.6%

Active Loads

18

Pending Sync

2

Scan Scrap Now

AI Scrap Scanner

Camera

Gallery

SPCB Compliance Hub

Detected Region

Kanpur, Uttar Pradesh

UPPCB

Regional Matrix

UPPCB

Applicable Rules

UPPCB EPR 2023

Permitted Materials

Steel, Al, Cu, Plastic, E-Waste

View Full Matrix

Sync Status

You are offline

Working in offline mode

Pending Records

18

Scans

12

Loads

4

Transactions

2

Last Sync

2h 15m ago

Sync Now

Generate QR

Load ID

LOAD-2024-05-18-001

Expires in

14:52

Share QR

Home

Scan

Loads

Profile

Field-Ready Production App running on Android Studio Emulator.

Native Camera AI Scanner for sub-second field scrap grading.

Location-Aware SPCB Compliance Hub detection (UPPCB/MPCB).

Offline-First SQLite cache with automated background cloud sync.

Multilingual Indic Voice Guidance for scrap yard operators.

Dynamic QR Code generation for zero-knowledge proof-of-transfer.

Sleek Onyx Neutral Dark & Light Studio Interface.

Live Circular Scrap Marketplace with real-time purity & carbon filters.

CPCB Statutory EPR Calculator with UPPCB/MPCB Regional Matrix.

Interactive AI Contamination Heatmap with visual bounding boxes.

Polygon Amoy Explorer with 1-click on-chain TX verification.

One-Click CPCB Form 1 PDF & JSON Government Export.

TECHNOLOGY
FOUNDATION

ML / AI MODEL

- Tool-calling LLM
- BGE-M3 embeddings
- Simulation + optimization engines

DATABASE & STORAGE

- PostgreSQL / PostGIS
- Redis event bus
- Local SQLite / Room cache
- Object storage (S3 / IPFS Backup)

CLOUD / EDGE

- Cloud APIs + processing
- Offline mobile layer
- Software-first design

APIs & CONNECTIONS

- Inputs → Ingestion → Digital Twin
- Route, Shelter, SOS & Logistics modules
- Outputs → Role-based operating apps

Citizen

Responder

Shelter /
Hospital

Command

ROLE-BASED INTERFACES

BUSINESS IMPACT, METRICS & SCALABILITY ROADMAP



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85%+

EPR Audit Time Saved

From 45 days manual paperwork to sub-minute automated verification



100%

Fraud Elimination

Zero ghost invoicing with on-chain dual handshake



₹ ₹15,000/MT

Penalties Avoided

Maximum statutory savings under EPA Section 15



3.24 kg

CO₂e Abated / kg

EPA WARM verified carbon reduction factor for recycled metals

PAN-INDIA SCALABILITY ROADMAP

01

Phase 1: Industrial Cluster Pilot (Q1 2026)



Onboard 50+ tier-1 manufacturing ancillaries and 20 certified recyclers in Noida (UPPCB) and Pune (MPCB).



02

Phase 2: SPCB & PRO API Integration (Q2-Q3 2026)



Direct integration with State Pollution Control Boards and Producer Responsibility Organizations.



03

Phase 3: National Carbon Marketplace (Q4 2026+)



Pan-India rollout supporting 10,000+ informal aggregators with micro-incentives & corporate Scope 3 offset trading.



Regulatory Compliance at Scale

Fully aligned with EPR Rules, CPCB & SPCB mandates.



Environmental Impact

Measurable CO₂e reduction with verified reporting.



Economic Value Creation

Cost savings, penalty avoidance & new carbon revenue streams.



Scalable & Interoperable Infrastructure

Built for nationwide adoption & future global expansion.



THANK YOU!

CIRCULARCHAIN — Autonomous Circular Economy & Carbon Ledger Protocol

“

Automating India's Industrial Scrap Lifecycle with
Multi-Modal AI, CPCB Statutory Compliance & Polygon Blockchain.

”



SUBMISSION RESOURCES & LIVE VERIFICATIONS



GitHub Repository

<https://github.com/CodewithEvilxd/Automate-India>



Mobile App

Flutter APK on
Android Studio Emulator (Pixel 6)



Pixel 6



TEAM MEMBERS

01

Nishant Gaurav

Team Lead · System Architecture & Web3

02

Rudra Pratap Nagar

AI / ML Intelligence & Computer Vision

03

Neeraj Kumar

Full-Stack Systems & Statutory Compliance



TEAM
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EVENT
AUTOMATE INDIA 2026



QUESTIONS &
LIVE DEMONSTRATION

